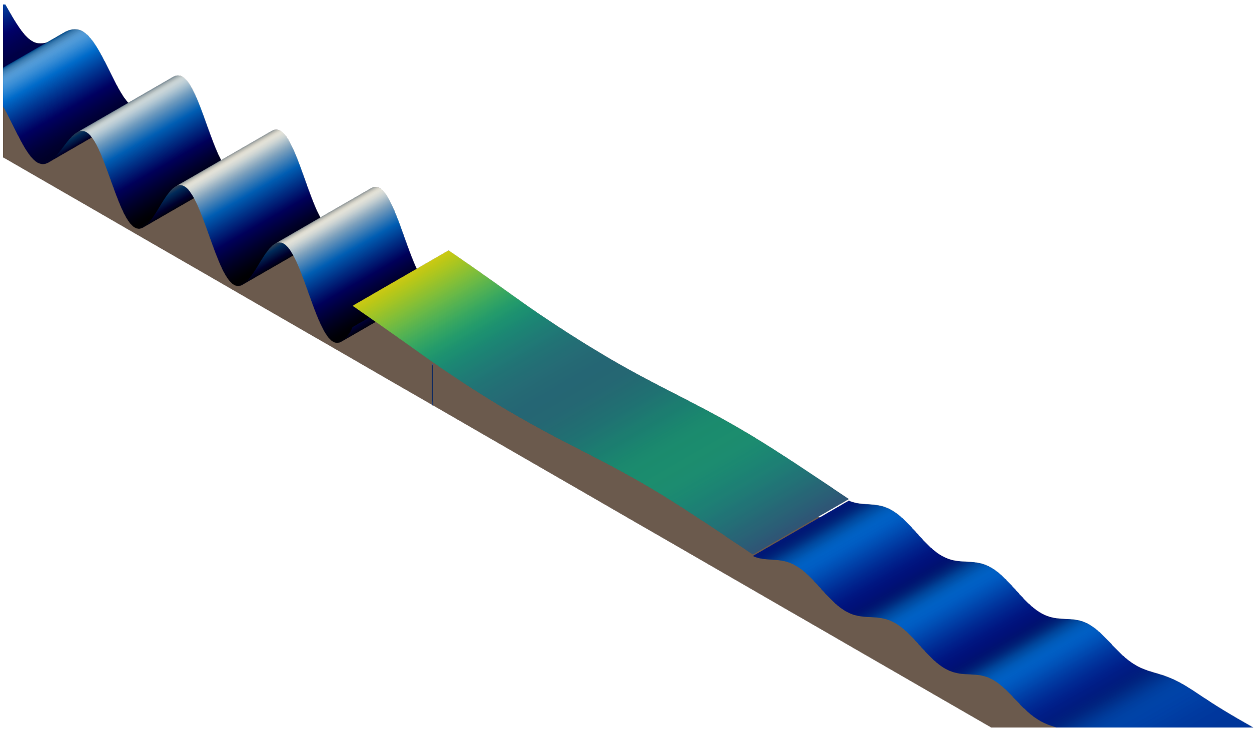
Hydroelastic Analysis of a Very Large Floating Structure Connected to the Sea Bottom by an Elastic Spring

ADVANCED COMPUTATIONAL MECHANICS

CIEM1301



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1 Introduction

Very Large Floating Structures (VLFS) in offshore environments are of increasing relevance in research. They can be found in various forms like photovoltaic structures, floating airports and floating islands [3]. Thus, the behaviour of these need to be studied in order to better understand and design such VLFS. VLFS are relatively thin in comparison to the other dimensions and would exhibit low stiffness. Due their their large size, the incoming waves are considered to be short. They are generally modelled as very thin plates where the response is a coupled problem between the plates and the fluid over which it floats. The plates can also be connected to the bed with an elastic member or members at different points in order to reduce the hydroelastic response of the VLFS.

A hydroelastic analysis of VLFS using a monolithic finite element formulation is discussed in detail by Colomés et al. [2]. They analyse VLFS of different shapes using the energy conservation formulation and therby, achieving a non-iterative algorithm. The structure is modelled using the Euler-Bernoulli beam and Poisson–Kirchhoff plate equations and the waves are described by the linear potential and free-flow theory. The linearity allows for both time and frequency domain analyses. A monolithic approach is implemented, where the governing equations are coupled and is based on the monolithic FEM method proposed by Akkerman et al. [1]. The finite element library Gridap is implemented in Julia for the computation of the numerical equations established. The formulations are based on continuous/discontinuous Galerkin approach where the discrete functions are continuous at the nodes but the gradient is discontinuous. For the frequency domain analysis, due to the Airy wave theory, the full discrete problem is reduced to a set of time independent and complex vale fields. Alternatively, for time domain analysis, the Newmark-beta time discretisation is adopted where two coefficients determine the stability and accuracy of the scheme. The error from the computation are compared for different element sizes and element orders, different time steps, and energy conservation. The results obtained are in good agreement with other works in literature as compared by them.

When dealing with VLFS for practical implementation, the reduction or control of vibrations are also of importance as it has direct a direct effect on the structural design. The elastic connector can consist of spring and / or dashpot connectors linking a corner or edge to the bottom of the sea bed. The characteristics of the connector can be tuned on the basis of the wave characteristics, in order to minimise or obtain the desired hydroelastic response. Khabakhpasheva and Korobkin [5] analysed the hydroelastic behaviour of compound floating plate in waves where they proposed two manners to reduce the vibrations of the VLFS. The first approach is by introduction of an auxiliary plate with its own stiffness properties, which provides reduction of the main structure deflections. The second approach connects the floating beam with the sea bottom using a spring, the stiffness properties of which can be adjusted such that the vibrations induced by the waves are reduced. Karperaki et al. [4] also analysed the transient reponse of a VLFS elastically connected to the sea bed with stiffness and dashpot connectors. Yet, the plate is modelled as a beam or a strip and applicable for slender VLFS. Whereas, Ngyuen et al. [6] modelled the plate as a Mindlin plate connected to the seabed by vertical elastic mooring lines at the edges in different configurations but only for the vertical movement control while the horizontal restraint is applied using mooring dolphin rudder-fender system. A hybrid finite element - boundary element method is adopted for the frequency domain hydroelastic analysis. The study examined the effectiveness of the mooring lines in reducing the hydroelastic responses for various wavelengths, incident wave angles and aspect ratios of the plate. The stiffnesses of the mooring lines are also optimised by employing the Differential Evolution Algorithm. It was found that the response is significantly reduced by the use of mooring lines.

In this report, a VLFS is hydroelastically analysed on the basis of the monolithic model presented by Colomés et al. [2], but with an elastic spring support on the plate edge connecting it to the bottom of the sea bed.

2 Problem Setting

Let us consider a the plate on a fluid domain Ω. The Z axis is assumed vertically upwards positive. The fluid is bounded by Γ_b at the bottom, Γ_{fs} at the fluid surface, Γ_{in} at the inlet and Γ_{out} at the outlet. The floating structure is bounded by $\Lambda_{\text{fs,str,1}}$ to $\Lambda_{\text{fs,str,2}}$. The plate is of length L and is connected to the seabed at the left edge $\Lambda_{\text{fs,str,1}}$ by a spring of stiffness k_s. The outward unit normal to the domain boundary is represented by **n**. The surface elevation is represented by η. The scheme is illustrated in figure 1.
The following assumptions are made:

1. The fluid is inviscid, incompressible and irrotational and can be described well by (linear) potential flow. It is also assumed that there is no cavitation, that is, detachment of the structure with respect to the fluid.

2. The incoming waves have a small steepness and can be well described by (linear) Airy wave theory.

3. The floating structures are thin and can be modelled by the linear Euler–Bernoulli equations.
4. The beam is assumed to be fixed in the horizontal direction.

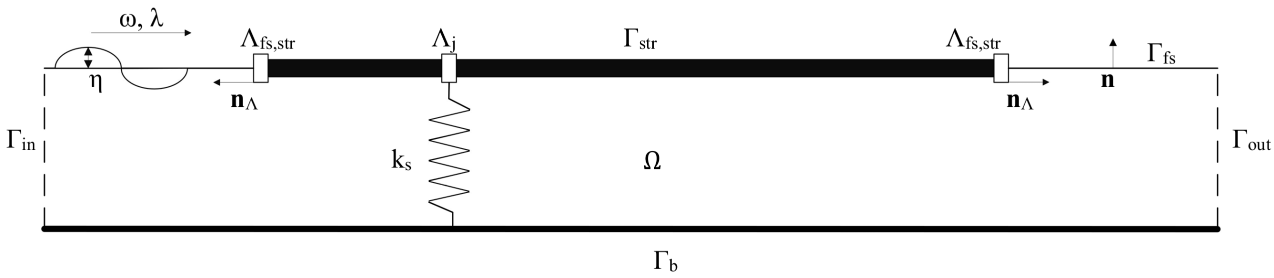

Figure 1: Sketch of the scheme with geometrical entities.

2.1 Linear Potential Flow

A scalar potential flow exists for the irrotational and inviscid flow with the velocity vector **u** in the fluid domain Ω such that **u** $= \nabla\phi$. Further, the incompressible flow assumption provides us with $\nabla \cdot$ **u** $= 0$, thus providing the fluid flow governing equation as follows.

$$\Delta\phi = 0 \quad \text{in } \Omega. \tag{1}$$

The kinematic and dynamic boundary conditions are as follows.

$$\mathbf{n} \cdot \nabla\phi = 0 \quad \text{on} \quad \Gamma_b, \tag{2}$$

$$\mathbf{n} \cdot \nabla\phi = u_{\text{in}} \quad \text{on} \quad \Gamma_{\text{in}}, \tag{3}$$

$$\mathbf{n} \cdot \nabla\phi = u_{\text{out}} \quad \text{on} \quad \Gamma_{\text{out}}, \tag{4}$$

$$\mathbf{n} \cdot \nabla\phi = \eta_t \quad \text{on} \quad \Gamma_{\text{fs}} \cup \Gamma_{\text{str}}. \tag{5}$$

Equation 2 enforces no penetration through the bottom bed, equations 3 and 4 prescribe an inlet and outlet normal velocities, and based on the second assumption the free surface and beam interface is described in equation 5. The linearised Bernoulli equation gives the pressure p at free surface with ρ_w as the fluid density and the gravity acceleration $g = 9.81 \text{ m/s}^2$, is as follows.

$$p = -\rho_w\phi_t - g\rho_w\eta - \frac{1}{2}(\nabla\phi)^2 \approx -\rho_w\phi_t - g\rho_w\eta \quad \Gamma_{\text{fs}} \cup \Gamma_{\text{str}} \tag{6}$$

2.2 Euler-Bernoulli Beam

Based on the third assumption, we have that the surface elevation is equivalent to the beam deflection at the contact surface.

$$\rho_b h_b \eta_{tt} + D\Delta^2\eta = p \quad \text{on } \Gamma_{\text{str}}. \tag{7}$$

Where, ρ_b is the structure density, h_b is the structure thickness and $D = EI$ is the constant structural rigidity. For the free surface, the pressure is atmospheric pressure, that is, $p = p_a = 0$, which gives us the following conditions.

$$\phi_t = g\eta \quad \text{on } \Gamma_{\text{fs}} \tag{8}$$

$$d_0\eta_{tt} + D_\rho\delta^2\eta + \phi_t + g\eta = 0 \quad \text{on } \Gamma_{\text{str}} \tag{9}$$

where, $d_0 = \frac{\rho_b h_b}{\rho_w}$ is the draft of the structure and $D_\rho = \frac{D}{\rho_w}$.
We consider the structure to have free motion at the boundaries. At the point where the spring is connected to the structure at Λ_{j}, there is a jump in shear force. This gives the boundary conditions and interface conditions for the structure as follows.

$$D\Delta\eta = 0 \quad \text{at } \Lambda_{\text{fs,str}} \tag{10}$$

$$\nabla(D\Delta\eta) \cdot \mathbf{n}_\Lambda = 0 \quad \text{at } \Lambda_{\text{fs,str}} \tag{11}$$

$$\nabla\langle D\Delta\eta\rangle \cdot \mathbf{n}_\Lambda = k_s\eta \quad \text{at } \Lambda_{\text{j}} \tag{12}$$

3 Finite Element Formulation

3.1 Weak form

In order to express the weak form of the governing equations, notations are introduced for compactness.

$$(u, v)_\Omega = \int_\Omega u(\mathbf{x})\ v(\mathbf{x}) d\Omega \tag{13}$$

$$[\![v]\!] = v^+ - v^- \quad \text{and} \quad \langle v \rangle = \frac{1}{2}\left(v^+ + v^-\right) \tag{14}$$

Using the governing equations from the previous section, the weak form is obtained as follows.

$$B_{EB}\left([\phi,\eta],[w,v]\right) = L([w,v]) \tag{15}$$

$$\begin{aligned} B_{EB}\left([\phi,\eta],[w,v]\right) = {} & (\nabla\phi,\nabla w)_\Omega - (\eta_t, w)_{\Gamma_{\mathsf{fs}}\cup\Gamma_{\mathsf{str}}} \\ & + \beta(\phi_t + g\eta, \alpha_f w + v)_{\Gamma_{\mathsf{fs}}} + (d_0\eta_{tt} + \phi_t + g\eta, v)_{\Gamma_{\mathsf{str}}} \\ & + (D_\rho\Delta\eta, \Delta v)_{\Gamma_{\mathsf{str}}} + (D_\rho k_s\langle\eta\rangle, \langle v\rangle)_{\Lambda_{\mathsf{j}}} \end{aligned} \tag{16}$$

$$L([w,v]) = (u_{\mathsf{in}}, w)_{\Gamma_{\mathsf{in}}} + (u_{\mathsf{out}}, w)_{\Gamma_{\mathsf{out}}} \tag{17}$$

Where α_f and β are scaling parameters introduced for stability and dimensional consistency purposes.

3.2 Spatial Discretisation

We can denote the spatial discretisation of domains using $(.)_h$ notation. The discretised bilinear and linear forms, based on the Galerkin FE formulation are given as follows.

$$\hat{B}_{h,EB}\left([\phi_h,\eta_h],[w_h,v_h]\right) = L([w_h,v_h]) \tag{18}$$

$$\begin{aligned} \hat{B}_{h,EB}\left([\phi_h,\eta_h],[w_h,v_h]\right) = {} & (\nabla\phi_h,\nabla w_h)_{\Omega_h} - (\eta_{h,t}, w_h)_{\Gamma_{\mathsf{fs,h}}\cup\Gamma_{\mathsf{str,h}}} \\ & + \beta(\phi_{h,t} + g\eta_h, \alpha_f w_h + v_h)_{\Gamma_{\mathsf{fs,h}}} + (d_0\eta_{h,tt} + \phi_{h,t} + g\eta_h, v_h)_{\Gamma_{\mathsf{str,h}}} \\ & + (D_\rho\Delta\eta_h, \Delta v_h)_{\Gamma_{\mathsf{str,h}}} + (D_\rho k_s\langle\eta_h\rangle, \langle v_h\rangle)_{\Lambda_{\mathsf{j,h}}} \\ & - \left(\langle D_\rho\Delta\eta_h\rangle, [\![\nabla v_h\cdot\mathbf{n}_\Lambda]\!]\right)_{\Lambda_{\mathsf{str,h}}} - \left([\![\nabla\eta_h\cdot\mathbf{n}_\Lambda]\!], \langle D_\rho\Delta v_h\rangle\right)_{\Lambda_{\mathsf{str,h}}} \\ & - \left(\langle D_\rho\Delta\eta_h\rangle, [\![\nabla v_h\cdot\mathbf{n}_\Lambda]\!]\right)_{\Lambda_{\mathsf{j,h}}} - \left([\![\nabla\eta_h\cdot\mathbf{n}_\Lambda]\!], \langle D_\rho\Delta v_h\rangle\right)_{\Lambda_{\mathsf{j,h}}} \\ & + \frac{\gamma}{h}([\![D_\rho\nabla\eta\cdot\mathbf{n}_\Lambda]\!], [\![\nabla v_h\cdot\mathbf{n}_\Lambda]\!])_{\Lambda_{\mathsf{str,h}}} + \frac{\gamma}{h}([\![D_\rho\nabla\eta\cdot\mathbf{n}_\Lambda]\!], [\![\nabla v_h\cdot\mathbf{n}_\Lambda]\!])_{\Lambda_{\mathsf{j,h}}} \end{aligned} \tag{19}$$

The last term in equation 19 is added for stability purposes and while the three terms before it are for the continuity of moment and shear in the beam.

3.3 Time Discretisation

Equation 19 is discrete in space and continuous second order differential in time. We can apply frequency domain analysis or time domain analysis for the solution.

3.3.1 Frequency Domain

The analysis based on Airy wave theory allows for frequency domain analysis where the response is assumed to be harmonic. The space-time dependent variable $\xi(\mathbf{x},t)$ can be expressed in terms of frequency ω and a time independent variable $\bar{\xi}$ as follows.

$$\xi(\mathbf{x},t) = \bar{\xi}(\mathbf{x})\exp(-i\omega t) \tag{20}$$

This applies to our equations as well, giving us time independent variables $\bar{\phi}(\mathbf{x})$ and $\bar{\eta}(\mathbf{x})$ but for simplicity in expression, only in the frequency domain analysis section, $\bar{\phi} = \phi$ and $\bar{\eta} = \eta$. Thus, we obtain the discrete equation as follows.

$$\hat{B}^{\omega}_{h,\mathrm{EB}}\left([\phi_h,\eta_h],[w_h,v_h]\right) = L^{\omega}([w_h,v_h]) \tag{21}$$

$$
\begin{aligned}
\hat{B}^{\omega}_{h,\mathrm{EB}}\left(\left[\phi_h,\eta_h\right],\left[w_h,v_h\right]\right) = {} & \left(\nabla\phi_h,\nabla w_h\right)_{\Omega_h} + \left(i\omega\eta_h, w_h\right)_{\Gamma_{\mathsf{fs},h}\cup\Gamma_{\mathsf{str},h}} \\
& + \beta\left(g\eta_h - i\omega\phi_h, \alpha^{\omega}_f w_h + v_h\right)_{\Gamma_{\mathsf{fs},h}} + \left((g-\omega^2 d_0)\eta_h - i\omega\phi_h, v_h\right)_{\Gamma_{\mathsf{str},h}} \\
& + \left(D_\rho\Delta\eta_h,\Delta v_h\right)_{\Gamma_{\mathsf{str},h}} + \left(D_\rho k_s\langle\eta_h\rangle,\langle v_h\rangle\right)_{\Lambda_{\mathsf{j},h}} \\
& - \left(\langle D_\rho\Delta\eta_h\rangle, [\![\nabla v_h\cdot\mathbf{n}_{\Lambda_{\mathsf{str},h}}]\!]\right)_{\Lambda_{\mathsf{str},h}} - \left([\![D_\rho\nabla\eta_h\cdot\mathbf{n}_{\Lambda_{\mathsf{str},h}}]\!],\langle\Delta v_h\rangle\right)_{\Lambda_{\mathsf{str},h}} \\
& - \left(\langle D_\rho\Delta\eta_h\rangle, [\![\nabla v_h\cdot\mathbf{n}_{\Lambda_{\mathsf{j},h}}]\!]\right)_{\Lambda_{\mathsf{j},h}} - \left([\![D_\rho\nabla\eta_h\cdot\mathbf{n}_{\Lambda_{\mathsf{j},h}}]\!],\langle\Delta v_h\rangle\right)_{\Lambda_{\mathsf{j},h}} \\
& + \frac{\gamma}{h}\left([\![D_\rho\nabla\eta_h\cdot\mathbf{n}_{\Lambda_{\mathsf{str},h}}]\!],[\![\nabla v_h\cdot\mathbf{n}_{\Lambda_{\mathsf{str},h}}]\!]\right)_{\Lambda_{\mathsf{str},h}} \\
& + \frac{\gamma}{h}\left([\![D_\rho\nabla\eta_h\cdot\mathbf{n}_{\Lambda_{\mathsf{j},h}}]\!],[\![\nabla v_h\cdot\mathbf{n}_{\Lambda_{\mathsf{j},h}}]\!]\right)_{\Lambda_{\mathsf{j},h}}
\end{aligned}
\tag{22}
$$

3.3.2 Time Domain

Instead of assuming a harmonic response, we can discretise the time domain for equation 18. The Newmark-beta time discretisation scheme can be used for the second order time ODE. Let us consider a constant time step Δt. The first and second time derivatives, as required, are as follows.

$$
x_t^{n+1} = \delta_t\left(x^{n+1} - x^n\right) + \frac{1-\gamma_{\mathsf{NB}}}{\beta_{\mathsf{NB}}}x_t^n + \Delta t\,\frac{1-\gamma_{\mathsf{NB}}}{2\beta_{\mathsf{NB}}}x_{tt}^n, \tag{23}
$$

$$
x_{tt}^{n+1} = \delta_{tt}\left(x^{n+1} - x^n\right) - \frac{1}{\beta_{\mathsf{NB}}\Delta t}x_t^n + \frac{1-2\beta_{\mathsf{NB}}}{2\beta_{\mathsf{NB}}}x_{tt}^n, \tag{24}
$$

where, $\delta_t = \frac{\gamma_{\mathsf{NB}}}{\beta_{\mathsf{NB}}\Delta t}$, $\delta_{tt} = \frac{1}{\beta_{\mathsf{NB}}\Delta t^2}$, $\gamma_{\mathsf{NB}} = 0.5$ and $\beta_{\mathsf{NB}} = 0.25$. γ_{NB} and β_{NB} are the coefficients which determine the stability and accuracy of the scheme.
Using the time discretisation schemes, we can apply it on 18 to obtain the following.

$$
\widehat{B}^{\omega}_{h,\mathrm{EB}}\big([\phi_h,\eta_h],[w_h,v_h]\big) = L^{\omega}_h\big([w_h,v_h]\big) \tag{25}
$$

$$
\begin{aligned}
\widehat{B}^{\omega}_{h,\mathrm{EB}}\big([\phi_h,\eta_h],[w_h,v_h]\big) = {} & (\nabla\phi_h,\nabla w_h)_{\Omega_h} - (\eta_{h,t}, w_h)_{\Gamma_{\mathsf{fs},h}\cup\Gamma_{\mathsf{str},h}} \\
& + \beta\big(\phi_{h,t} + g\eta_h, \alpha_f w_h + v_h\big)_{\Gamma_{\mathsf{fs},h}} + (d_0\eta_{h,tt} + \phi_{h,t} + g\eta_h, v_h)_{\Gamma_{\mathsf{str},h}} \\
& + (D_\rho\Delta\eta_h,\Delta v_h)_{\Gamma_{\mathsf{str},h}} + (D_\rho k_s\langle\eta_h\rangle,\langle v_h\rangle)_{\Lambda_{\mathsf{j},h}} \\
& - \big(\langle D_\rho\Delta\eta_h\rangle,[\![\nabla v_h\cdot\mathbf{n}_{\Lambda_{\mathsf{str},h}}]\!]\big)_{\Lambda_{\mathsf{str},h}} - \big([\![\nabla\eta_h\cdot\mathbf{n}_{\Lambda_{\mathsf{str},h}}]\!],\langle D_\rho\Delta v_h\rangle\big)_{\Lambda_{\mathsf{str},h}} \\
& - \big(\langle D_\rho\Delta\eta_h\rangle,[\![\nabla v_h\cdot\mathbf{n}_{\Lambda_{\mathsf{j},h}}]\!]\big)_{\Lambda_{\mathsf{j},h}} - \big([\![\nabla\eta_h\cdot\mathbf{n}_{\Lambda_{\mathsf{j},h}}]\!],\langle D_\rho\Delta v_h\rangle\big)_{\Lambda_{\mathsf{j},h}} \\
& + \frac{\gamma}{h}\big([\![D_\rho\nabla\eta_h\cdot\mathbf{n}_{\Lambda_{\mathsf{str},h}}]\!],[\![\nabla v_h\cdot\mathbf{n}_{\Lambda_{\mathsf{str},h}}]\!]\big)_{\Lambda_{\mathsf{str},h}} \\
& + \frac{\gamma}{h}\big([\![D_\rho\nabla\eta_h\cdot\mathbf{n}_{\Lambda_{\mathsf{j},h}}]\!],[\![\nabla v_h\cdot\mathbf{n}_{\Lambda_{\mathsf{j},h}}]\!]\big)_{\Lambda_{\mathsf{j},h}}
\end{aligned}
\tag{26}
$$

4 Numerical Results

Let us consider a travelling wave as described by the following expressions.

$$
\mathbf{n}\cdot\nabla\phi = 0 \quad \text{on } \Gamma_b \tag{27}
$$

$$
\mathbf{n}\cdot\nabla\phi = -\omega\eta_0\frac{\cosh(k_\lambda y)}{\sinh(k_\lambda H)}\cos\left(k_\lambda x - \omega t\right) \quad \text{on } \Gamma_{\mathsf{in}} \tag{28}
$$

$$
\mathbf{n}\cdot\nabla\phi = 0 \quad \text{on } \Gamma_{\mathsf{out}} \tag{29}
$$

where k_λ is the wave number, ω the frequency. The dispersion equation is as follows.

$$
\omega = \sqrt{gk_\lambda\tanh(k_\lambda H)} \tag{30}
$$

$$
\lambda = \alpha L \tag{31}
$$

A damping zone at the inlet of the tank and at the outlet of the tank of length $L_d \approx 4L$ is defined and added to the free surface kinematic and dynamic boundary conditions as follows.

$$
\mathbf{n}\cdot\nabla\phi = \eta_t + \mu_2(\eta - \eta^*) \quad \text{on } \Gamma_{fs}, \tag{32}
$$

$$\phi_t + g\eta + \mu_1(\nabla\phi \cdot \mathbf{n} - \nabla\phi^* \cdot \mathbf{n}) = 0 \quad \text{on } \Gamma_{fs}. \tag{33}$$

$$\mu_1(x) = \begin{cases} \mu_0 \left[1 - \sin\left(\frac{\pi}{2}\frac{x}{L_d}\right)\right], & \text{if } x_{d,in} < x, \\ \mu_0 \left[1 - \cos\left(\frac{\pi}{2}\frac{x - x_d}{L_d}\right)\right], & \text{if } x > x_{d,out}, \\ 0, & \text{otherwise} \end{cases} \tag{34}$$

$$\mu_2(x) = k_\lambda \mu_1(x) \tag{35}$$

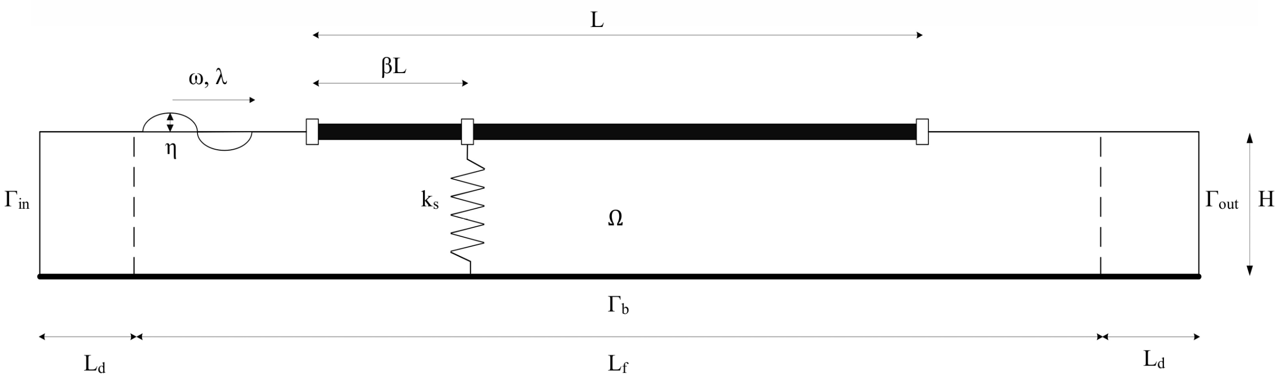

Figure 2: The geometry used for the analysis.

4.1 Sanity check of the effect of spring stiffness

Table 1: Parameters used for numerical analysis.

Parameter	Symbol	Value	Units
Draft	d_0	$8.1561 \cdot 10^{-3}$	m
Structure length	L	12.5	m
Fluid domain length	L_f	25	m
Tank depth	H	1.1	m
Connection location parameter	β	0.2	-
Spring stiffness	k_s	100	N/m
Structural rigidity	D_ρ	47100	N m
Gravity acceleration	g	9.81	m/s^2
Wavelength-to-beam length ratio	α	0.249	-

[h!]
The displacement profile results as seen in figures 3 and 4 appear to be in agreement. The portion of the plate before the spring shows higher displacement compared to the locations after the spring. There also appears to be no slope discontinuity in the beam. The 3D representation of the solution in time domain is represented in figure 5.

To further observe the effect of the spring, two more cases are considered with $k_s = 0$, and $k_s = \infty$ in the frequency domain analysis and can be seen in figure 6. The case with no spring stiffness shows a higher displacements along the entire plate with a higher value at the end. On the other hand, the rigid connection significantly affects the portion before the connection point and increases the response but lowers the response after it. The comparison of the spring stiffness cases are illustrated in figure 7. As expected from the rigid condition, there is no displacement at the point of connection. Furthermore, there appears to be a discontinuity in slope at the connection point but that is due to the nature of the absolute values being plotted. When investigating the real and imaginary values, there appears no such jump but rather an intersection of them.

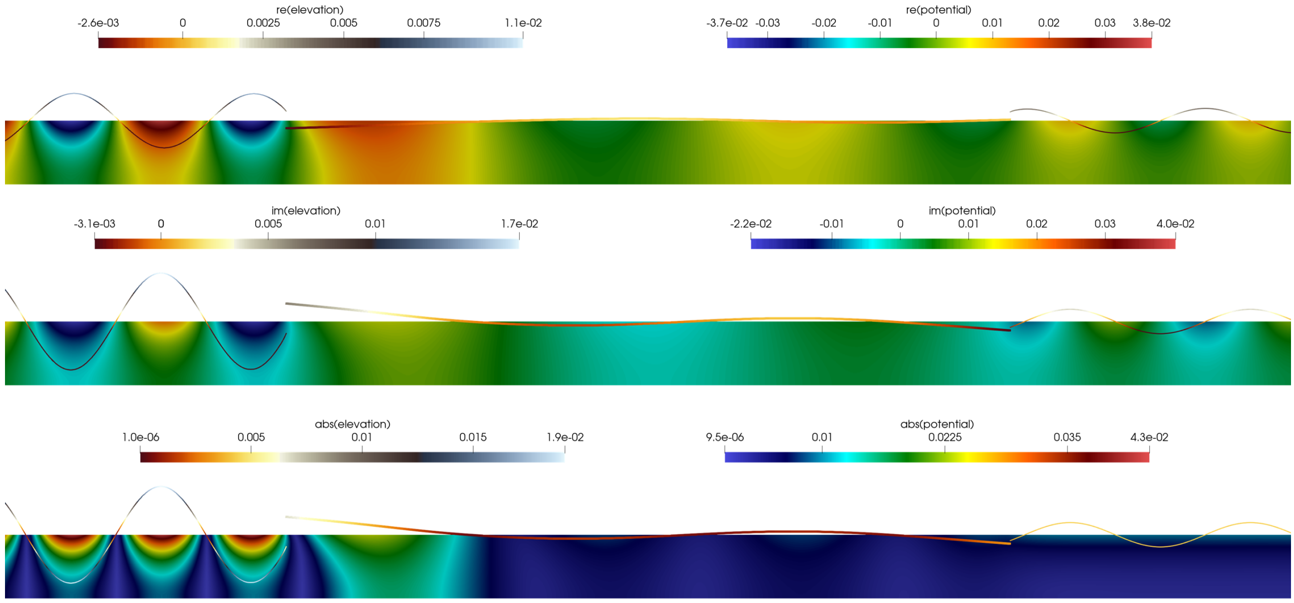

Figure 3: Velocity potential ϕ_h, and surface elevation η_h. Real part (top), imaginary part (centre), and absolute values (bottom). The surface elevation is scaled by 50.

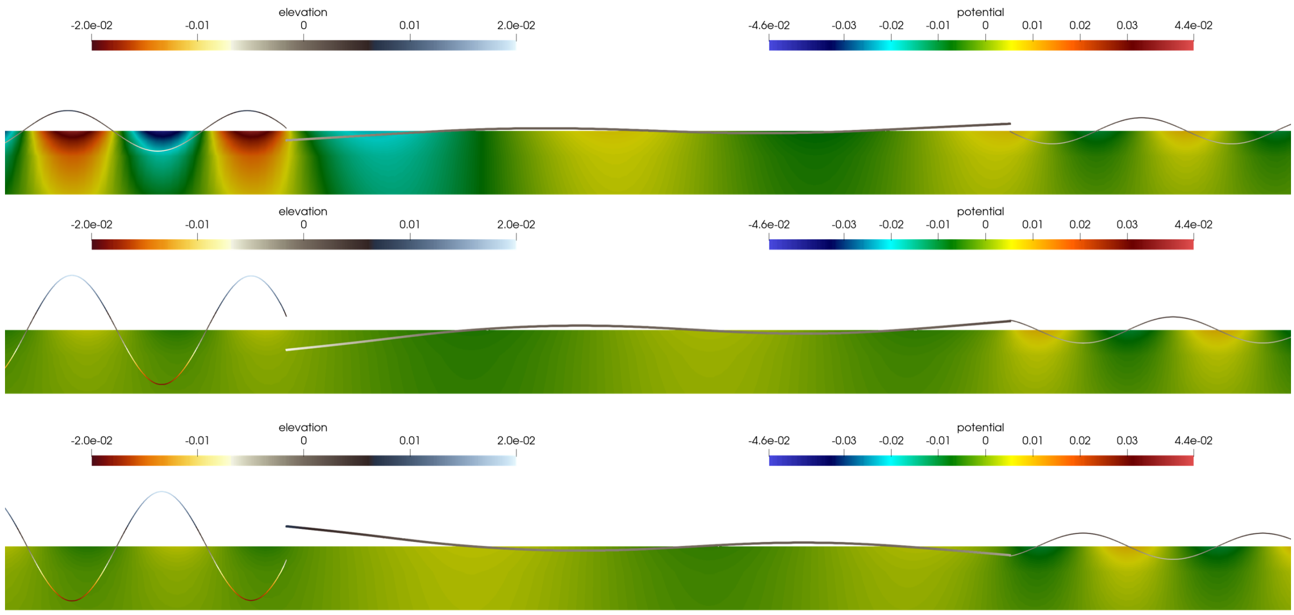

Figure 4: Velocity potential ϕ_h, and surface elevation η_h in m. $t = 25.22$ s (top), $t = 25.47$ (middle) and $t = 26.18$ s (bottom). The surface elevation is scaled by 50.

4.2 Effect of structural rigidity

Based on figure 7, it can be seen that the portion of the plate before the spring connection experiences higher response. To investigate the effect of structural rigidity D_ρ, two separate values are assigned as D_1 and D_2 for the portion before the spring connection and after respectively. The result is illustrated in figure 8. It can be seen that the minimum response can be observed for the case $D_1 = D_2/10$.

4.3 Comparison with rotational spring model

As mentioned in the introduction, Khabakhpasheva and Korobkin [5] proposed two methods for control of vertical vibration. One involving a vertical spring and the other involving an articulated plate. For the comparison, one of the models used by Colomés et al. [2], illustrated in figure 9, is used.
The response profile observed figure 10 shows similar shape for the second portion of the beam. The maximum response as well as the overall response of the beam is lower for the vertical spring case.

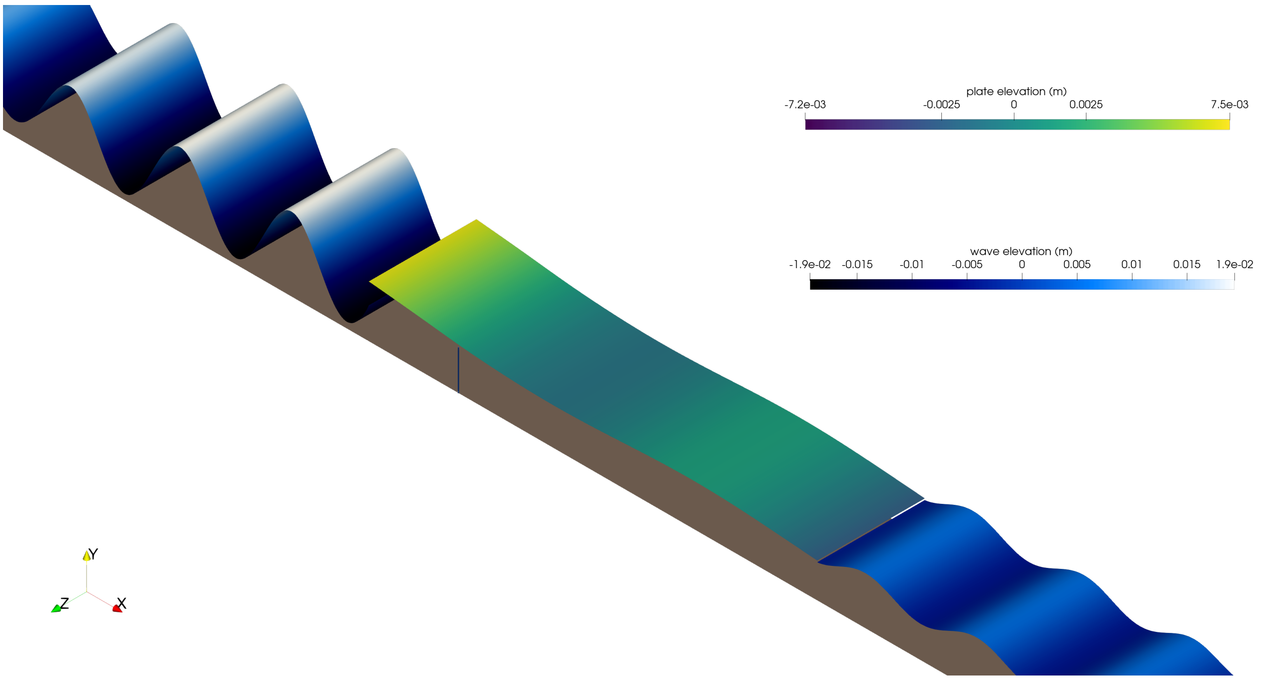

Figure 5: 3D representation of the model with plate, wave and mooring line linearly extruded for time domain analysis at $t = 23.35$ s.

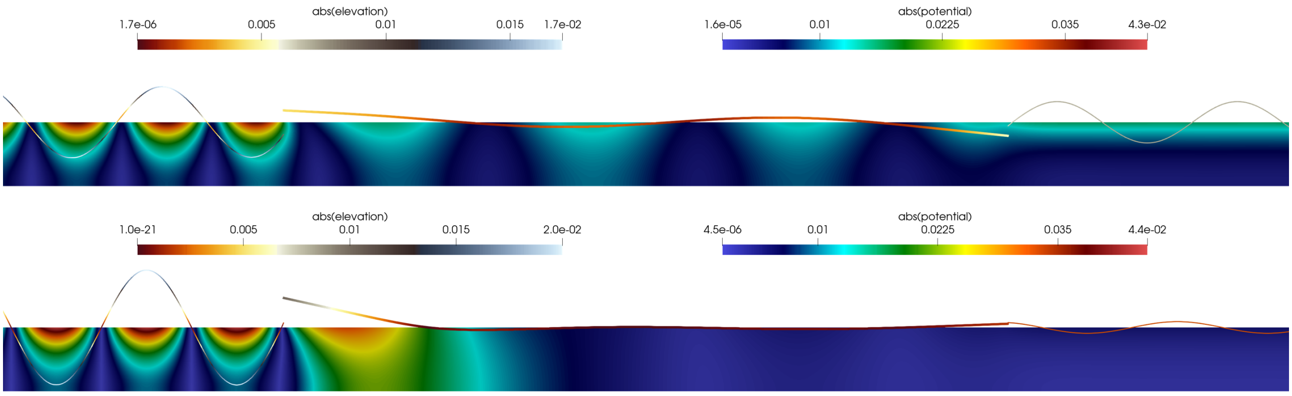

Figure 6: Velocity potential ϕ_h, and surface elevation η_h. $k_s = 0$ (top), and $k_s = \infty$ (bottom). The surface elevation is scaled by 50.

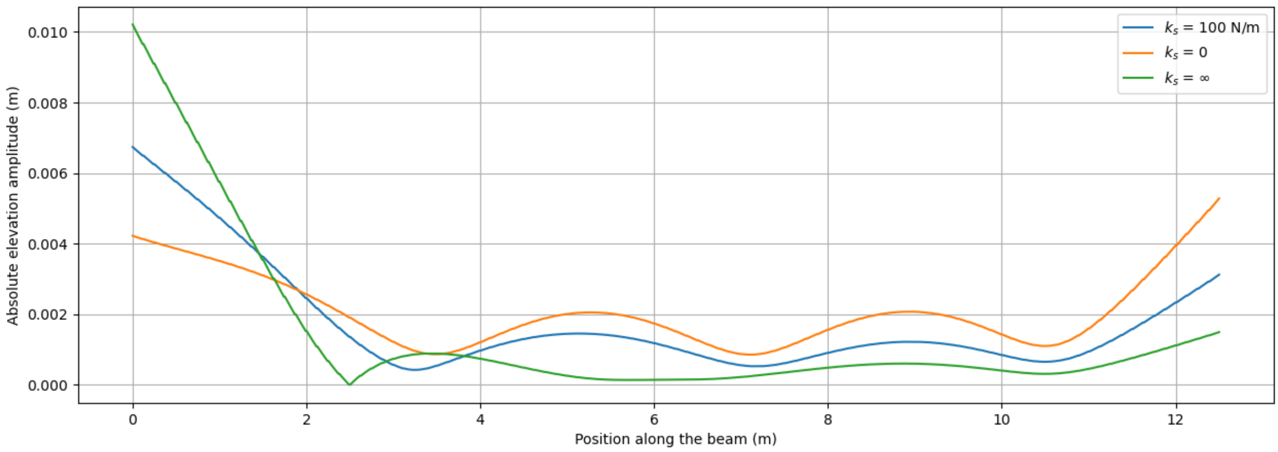

Figure 7: Absolute values of beam elevation amplitude η_h for different spring stiffness k_s.

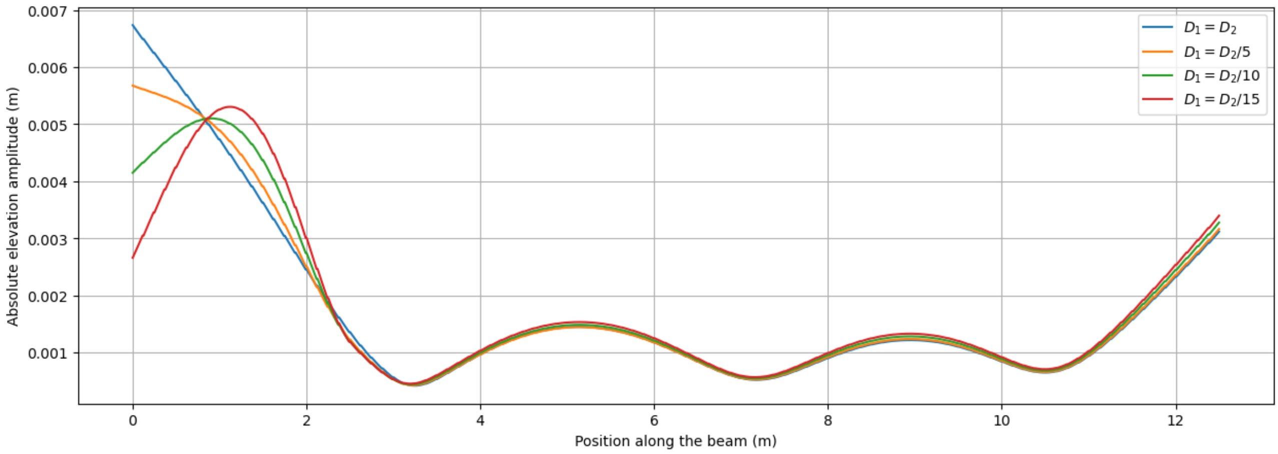

Figure 8: Absolute values of beam elevation amplitude η_h for different structural rigidities.

Table 2: Parameters used for comparison of vertical spring and rotational spring models.

Parameter	Symbol	Value	Units
Vertical spring stiffness	k_s	100	$\mathrm{N/m}$
Rotational spring stiffness	k_r	203.41	Nm
Structural rigidity	D_1	4710	$\mathrm{N\,m}$
Structural rigidity	D_1	47100	$\mathrm{N\,m}$

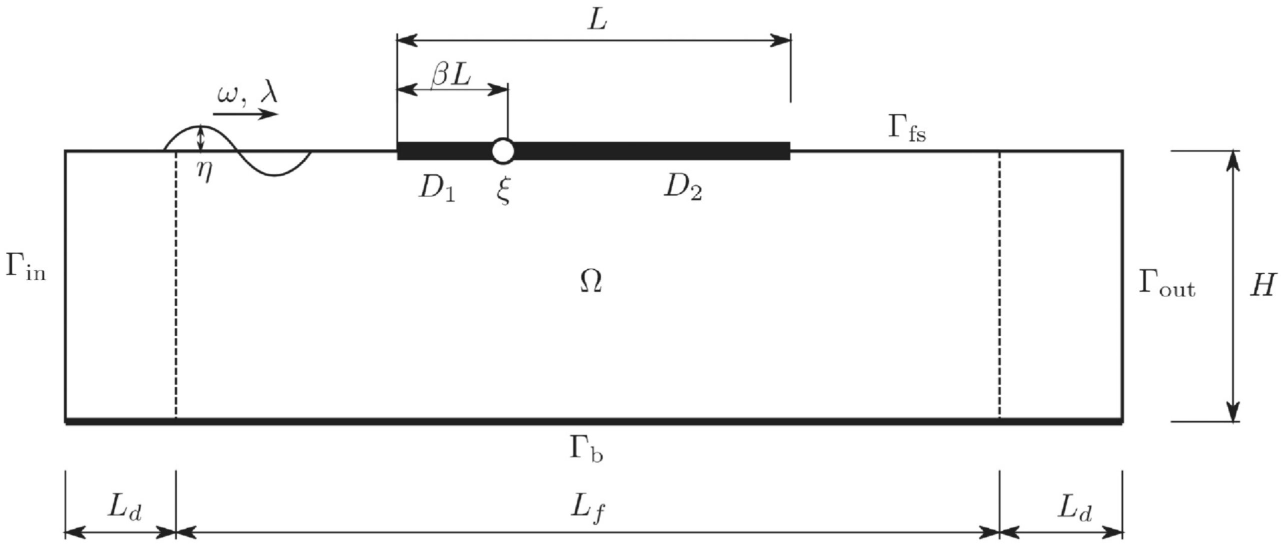

Figure 9: Illustration of the model with a rotational spring.

4.4 Effect of wave length

For all the previous analyses, the factor $\alpha = 0.249$, where α is defined in equation 31. The parameters for the structural rigidity and vertical spring stiffness are the same as described in table 2, while the α values are changed. It can be observed from figure 11, that lower α corresponds to smaller wave lengths forming in the structure which is expected. Yet, a smaller α does not necessarily mean smaller response as the crests and the edges can show higher responses as observed.

4.5 Effect of the location of the spring

In all the previous analyses, the location of the spring was based on $\beta = 0.2$. It can also be seen that the response before the spring is much higher than that after the spring. Figure 12 shows different the response for varying β values. Reduction from 0.2 to 0.1 leads to a high increase of the response at the front end while there is less reduction of the response after the connection point. This latter trend can be seen for the further reduction in β values.

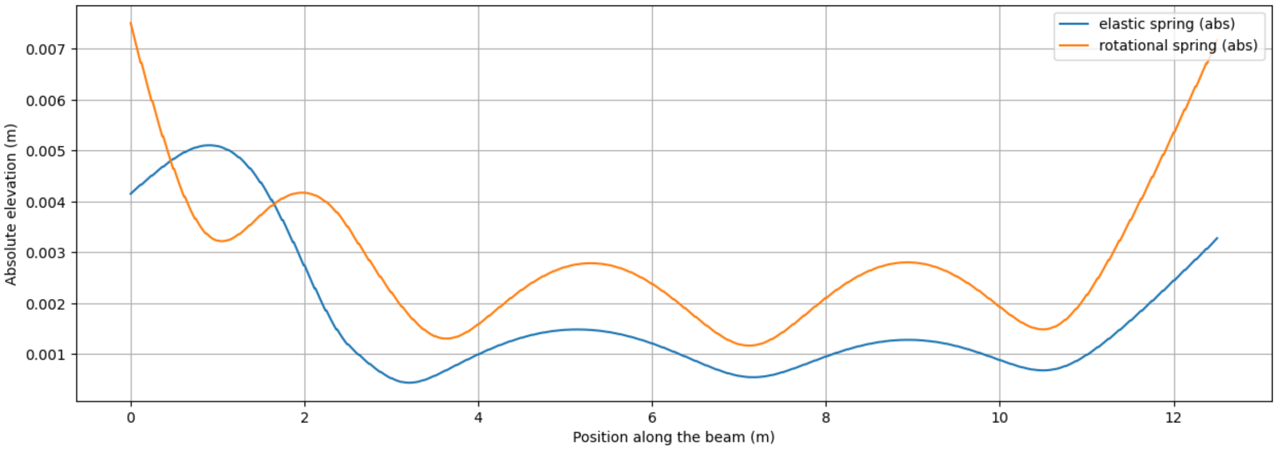

Figure 10: Absolute values of beam elevation amplitude η_h for vertical spring model and rotational spring model

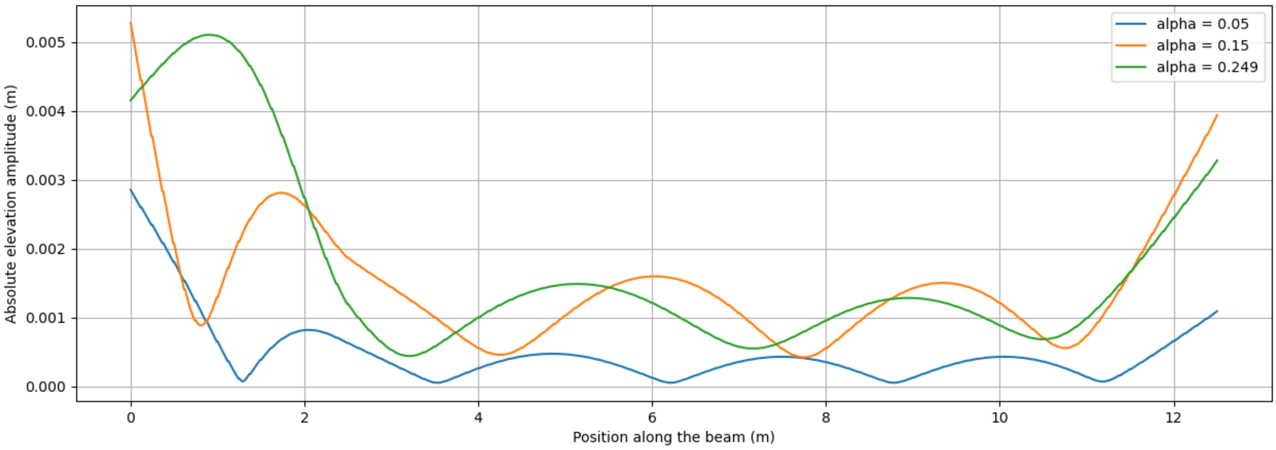

Figure 11: Absolute values of beam elevation amplitude η_h for different α values.

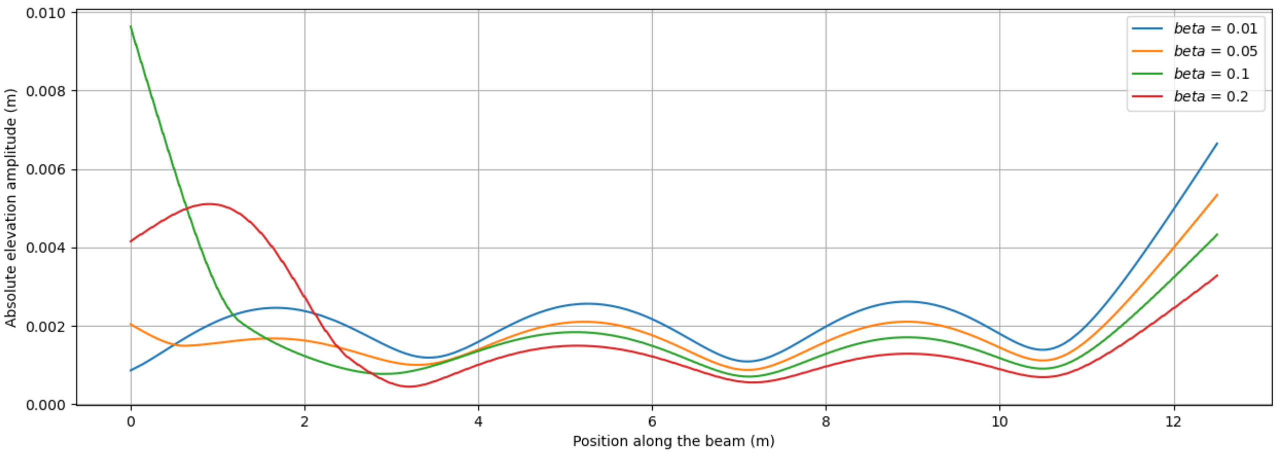

Figure 12: Absolute values of beam elevation amplitude η_h for different β values.

5 Conclusion

A Very Large Floating Structure (VLFS) connected to the seabed by a vertical spring is modelled using a monolithic finite element approach. The VLFS is generalised as an Euler-Bernoulli beam and the fluid is described by linear potential flow. A series of frequency domain and time domain analyses are performed for various parameters. A sanity check is performed by setting the spring stiffness values to the extremes and the result does lead to an expected profile for the cases set. Moreover, the effect of structural rigidity is studied by modifying the structural rigidity of the left portion of the beam before the spring connection point.

It is observed that lowering the structural rigidity can reduce the responses of the front portion but a slight increase in response for the rest of the structure. Additionally, the model is also compared with a model with rotational spring instead of a vertical spring at the same location. The vertical spring leads to lower responses along the entire profile of the beam. A direct comparison of the actual values of the stiffnesses themselves is could not be establish and thus, in practice, a comparison could be established. In addition to that, the effect of the wave length to the total length of the beam is investigated, from which, it can be seen that smaller ratio leads to leads to waves of smaller wave length appearing in the response but the peak amplitude may not necessarily be reduced. Moreover, the effect of the location of the vertical spring along the beam is analysed. It could be observed that a larger distance from the front edge leads to a lower response for the back portion of the beam but the front portion has no such simple relation and would require a closer analysis. The use of vertical mooring lines do reduce the vertical vibrations of the structure. Whether the front portion of the beam is to be treated as just a response reduction portion or to be actually be put to the use as that of the rest of the beam / plate depends on the practical scenarios and the required limits of vertical vibrations. In conclusion, the analyses performed in this report demonstrate the potential application of a VLFS moored vertically, represented by a spring, to the seabed based on a monolithic finite element approach.

6 Further Research

Due to time constraints, the scope of the report is very limited. In order to be able to implement VLFS with vertical moring lines in practice, further analyses would have to be performed. Following are a list of possible directions:

- Effect of wave frequency on the response by comparing with the natural frequency of the structure.
- Modelling the vertical mooring line as an elastic material to analyse the mooring line.
- Modelling an inclined mooring line while also modelling the horizontal movement of the beam / plate.
- Effect of a spring dashpot model for the mooring system.
- Effect of location of mooring lines for a 2D plate.
- Optimisation problems for the aforementioned effect of variables.

References

[1] Akkerman, I., Meijer, J., and ten Eikelder, M. (2020). Isogeometric analysis of linear free-surface potential flow. *Ocean Engineering*, 201:107114.

[2] Colomés, O., Verdugo, F., and Akkerman, I. (2023). A monolithic finite element formulation for the hydroelastic analysis of very large floating structures. *International Journal for Numerical Methods in Engineering*, 124(3):714–751.

[3] Gao, R., Tay, Z., Wang, C., and Koh, C. (2011). Hydroelastic response of very large floating structure with a flexible line connection. *Ocean Engineering*, 38(17):1957–1966.

[4] Karperaki, A., Belibassakis, K., and Papathanasiou, T. (2016). Time-domain, shallow-water hydroelastic analysis of vlfs elastically connected to the seabed. *Marine Structures*, 48:33–51.

[5] Khabakhpasheva, T. and Korobkin, A. (2002). Hydroelastic behaviour of compound floating plate in waves. *Journal of Engineering Mathematics*, 44(1):21–40.

[6] Nguyen, H., Dai, J., Wang, C., Ang, K., and Luong, V. (2018). Reducing hydroelastic responses of pontoon-type vlfs using vertical elastic mooring lines. *Marine Structures*, 59:251–270.

A Github Repository

The Github repository contains all the code used for the models and the generated graphs.
Link to the repository: github.com/Aqdas-M/Advanced-Computation-VLFS-with-Spring-